COMPANION FOR CHAPTER 7

OF

FUNDAMENTALS OF PROBABILITY WITH STOCHASTIC PROCESSES

FOURTH EDITION

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Chapter 7

Special Continuous Distributions

7A ADDITIONAL EXAMPLES

Example 7a In a measurement, a number is rounded off to the nearest k decimal places. Let X be the rounding error. Determine the distribution function of X and its parameters.

Solution: Round off error to the nearest integer is uniform over $(-0.5, 0.5)$; round off error to the nearest 1st decimal place is uniform over $(-0.05, 0.05)$; round off error to the nearest 2nd decimal place is uniform over $(-0.005, 0.005)$, and so on. In general, round off error to the nearest k decimal places is uniform over $(-5/10^{k+1},\ 5/10^{k+1})$. Therefore, the distribution function of X is

$$F(t) = \begin{cases} 0 & t < -5/10^{k+1} \\ \dfrac{t + \dfrac{5}{10^{k+1}}}{\dfrac{5}{10^{k+1}} + \dfrac{5}{10^{k+1}}} = 10^k t + \dfrac{1}{2} & -5/10^{k+1} \le t < 5/10^{k+1} \\ 1 & t \ge 5/10^{k+1}. \end{cases} \quad \blacklozenge$$

Example 7b As discussed in Section 4.5 of the book, for a random variable X, the true average magnitude of the fluctuations of X from $E(X)$ is $E\big(|X - E(X)|\big)$. However, since, mathematically, this quantity is difficult to handle, the quantity $\sqrt{E\big[(X - E(X))^2\big]}$ is used and is called the standard deviation of X. Let X be a normal random variable with mean μ and standard deviation σ. Find $E\big(|X - \mu|\big)$, the true average magnitude of the fluctuations of X from μ.

Solution: Clearly,

$$E\big(|X - \mu|\big) = \int_{-\infty}^{\infty} |x - \mu| \, \frac{1}{\sigma\sqrt{2\pi}} \exp\Big[\frac{-(x-\mu)^2}{2\sigma^2}\Big]\, dx.$$

Let $x - \mu = t$; then $dx = dt$, and

$$E\big(|X-\mu|\big) = \int_{-\infty}^{\infty} |t|\, \frac{1}{\sigma\sqrt{2\pi}} \exp\Big(\frac{-t^2}{2\sigma^2}\Big)\, dt.$$

Since the function $g(t) = \dfrac{1}{\sigma\sqrt{2\pi}}|t| \exp\Big(\dfrac{-t^2}{2\sigma^2}\Big)$ is an even function; that is $g(-t) = g(t)$, we have

$$E\big(|X-\mu|\big) = 2\int_{0}^{\infty} |t|\, \frac{1}{\sigma\sqrt{2\pi}} \exp\Big(\frac{-t^2}{2\sigma^2}\Big)\, dt = \frac{2}{\sigma\sqrt{2\pi}} \int_0^{\infty} t \exp\Big(\frac{-t^2}{2\sigma^2}\Big)\, dt.$$

By making the substitution $u = \dfrac{t^2}{2\sigma^2}$, we have $du = \dfrac{1}{\sigma^2}\, t\, dt$. So

$$E\big(|X-\mu|\big) = \frac{1}{\sigma}\sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-u}\sigma^2\, du = \sigma\sqrt{\frac{2}{\pi}}\Big[-e^{-u}\Big]_0^{\infty} = \sigma\sqrt{\frac{2}{\pi}}. \quad \blacklozenge$$

Example 7c Code 1 for "yes," and code 0 for "no" are transmitted through a noisy channel regularly. However, the quality of signals and data are degraded by the noise that the channel adds. Suppose that when a code X is sent, independently of other transmissions, another code $X + \epsilon$ is received, where $\epsilon \sim N(0, 1/16)$. If all the codes that are received and are less than or equal to 1/2 are interpreted as 0, and the ones that are received and are greater than 1/2 are interpreted as 1, what is the probability that the code 11001001 sent is decoded correctly when transmitted?

Solution: If the transmitted code sent is 0, when received, it will be interpreted as 0 with probability $P(\epsilon \le 1/2)$. If the transmitted code is 1, when received, it will be interpreted as 1 with probability $P(\epsilon > -1/2)$. Since $\epsilon \sim N(0, 1/16)$, we have

$$P\Big(\epsilon \le \frac{1}{2}\Big) = P\Big(Z \le \frac{1/2 - 0}{1/4}\Big) = P(Z \le 2) = \Phi(2) = 0.9772;$$

$$P\Big(\epsilon > -\frac{1}{2}\Big) = P\Big(Z > \frac{-1/2 - 0}{1/4}\Big) = P(Z > -2) = 1 - \Phi(-2) = \Phi(2) = 0.9772.$$

Therefore, when transmitted, each of the codes 0 and 1 is received and interpreted correctly with probability 0.9772, independently of other codes. Hence the probability that 11001001 is transmitted and interpreted as 11001001 is $(0.9772)^{8} \approx 0.83$. ♦

Example 7d At a prime location, due to the shortage of homes for sale, some home-buyers make offers above the asking prices. Morgan has put her house on the market and has decided to sell it to the first person who offers her at least \$350,000. If the offers are exponential random variables with mean \$320,000, independently of one another, on the average, how many offers will Morgan receive until she sells her house?

Solution: Call an offer a "success" if it is at least \$350,000. Let X be a random offer, in tens of thousands dollars. Then the probability of "success" is

$$p = P(X \ge 3.5) = \int_{3.5}^{\infty} \frac{1}{3.2} e^{-x/3.2}\, dx \approx 0.335.$$

We are interested in the expected value of the number of offers until the first "success." That is a geometric random variable with mean $\dfrac{1}{p} \approx \dfrac{1}{0.335} \approx 2.99.$ ♦

Example 7e The hazard rate of device A is given by

$$\lambda(t) = \begin{cases} t/10 & 0 < t < 10 \text{ years} \\ 10 & t \geq 10 \text{ years.} \end{cases}$$

(i) What is the probability that device A having lasted for 7 years lasts at least another 2 years?

(ii) The failure rate of device B is twice the rate at which A fails. What is the probability that device B having lasted 7 years lasts at least another 2 years?

Solution: Let X and Y be the lifetimes of devices A and B, respectively. Then using (7.6), the answer to (i) is

$$P(X \geq 9 \mid X \geq 7) = \frac{P(X \geq 9)}{P(X \geq 7)} = \frac{\exp\left(-\int_0^9 \frac{t}{10} dt\right)}{\exp\left(-\int_0^7 \frac{t}{10} dt\right)} \approx 0.202.$$

Similarly, the answer to (ii) is obtained as follows.

$$P(Y \geq 9 \mid Y \geq 7) = \frac{P(Y \geq 9)}{P(Y \geq 7)} = \frac{\exp\left(-\int_0^9 \frac{t}{5} dt\right)}{\exp\left(-\int_0^7 \frac{t}{5} dt\right)} \approx 0.041.$$

Therefore, device B has a much lower chance of surviving 2 or more years than device A. This is because device B has a failure rate that is much larger than the failure rate of A. ♦

Example 7f Let X be an exponential random variable with parameter $\theta > 0$. Let

$$Y = \begin{cases} X & \text{with probability } 1/2 \\ -X & \text{with probability } 1/2 \end{cases}$$

Find the probability density function of Y.

Solution: Let f and F be the density and distribution functions of X, respectively. Let g and G be the density and distribution functions of Y, respectively. By the law of total probability, we have

$$
\begin{aligned}
G(t) &= P(Y \le t) = P(Y \le t \mid Y = X)P(Y = X) + P(Y \le t \mid Y = -X)P(Y = -X) \\
&= P(X \le t) \cdot \frac{1}{2} + P(-X \le t) \cdot \frac{1}{2} \\
&= \begin{cases} \dfrac{1}{2}P(X \le t) + \dfrac{1}{2} & t \ge 0 \\[2ex] \dfrac{1}{2}P(X \ge -t) & t \le 0 \end{cases} \\
&= \begin{cases} \dfrac{1}{2}P(X \le t) + \dfrac{1}{2} & t \ge 0 \\[2ex] \dfrac{1}{2} - \dfrac{1}{2}P(X \le -t) & t \le 0 \end{cases} \\
&= \begin{cases} \dfrac{1}{2}F(t) + \dfrac{1}{2} & t \ge 0 \\[2ex] \dfrac{1}{2} - \dfrac{1}{2}F(-t) & t \le 0 \end{cases}
\end{aligned}
$$

Differentiating both sides of this relation, we obtain

$$
g(t) = \begin{cases} \dfrac{1}{2}f(t) & t \ge 0 \\[2ex] \dfrac{1}{2}f(-t) & t \le 0 \end{cases}
$$

Therefore,

$$
g(t) = \frac{1}{2}f(|t|) = \frac{1}{2}\theta e^{-\theta|t|} \qquad -\infty < t < \infty.
$$

So Y has Laplace distribution (See Exercise 16 of Section 7.3 of the book). ♦

Example 7g Show that

$$
F(t) = \begin{cases} 0 & t < 0 \\[2ex] \dfrac{2}{\pi}\arcsin(\sqrt{t}) & 0 \le t < 1 \\[2ex] 1 & t \ge 1 \end{cases}
$$

is the distribution function of a beta random variable with parameters $(1/2, 1/2)$.

Solution: Let f be the probability density function of a random variable X with distribution function F. Then

$$
f(t) = F'(t) = \begin{cases} \dfrac{2}{\pi} \cdot \dfrac{1}{2\sqrt{t}} \cdot \dfrac{1}{\sqrt{1 - (\sqrt{t})^2}} = \dfrac{1}{\pi\sqrt{t(1-t)}} & 0 < t < 1 \\[3ex] 0 & \text{otherwise.} \end{cases}
$$

Since, for $0 < t < 1$,

$$f(t) = \frac{1}{\sqrt{\pi}\sqrt{\pi}} t^{-1/2}(1-t)^{-1/2},$$

if we show that $\Gamma(1/2) = \sqrt{\pi}$, then we have

$$f(t) = \frac{\Gamma(1/2 + 1/2)}{\Gamma(1/2)\Gamma(1/2)} t^{-1/2}(1-t)^{-1/2} = \frac{1}{B(1/2, 1/2)}\, t^{-1/2}(1-t)^{-1/2},$$

which is the probability density function of a beta random variable with parameters $(1/2, 1/2)$. What is left is to show that $\Gamma(1/2) = \sqrt{\pi}$. Note that,

$$\Gamma(1/2) = \int_0^\infty t^{-1/2} e^{-t}\, dt.$$

Making the substitution $t = y^2/2$, we get

$$\begin{aligned}\Gamma\Big(\frac{1}{2}\Big) &= \sqrt{2}\int_0^\infty e^{-y^2/2}\, dy = \frac{\sqrt{2}}{2}\int_{-\infty}^\infty e^{-y^2/2}\, dy \\ &= \sqrt{\pi} \cdot \frac{1}{\sqrt{2\pi}}\int_{-\infty}^\infty e^{-y^2/2}\, dy = \sqrt{\pi}. \quad \blacklozenge\end{aligned}$$

7B BLACK-SCHOLES CALL OPTION FORMULA

To show the wide range of applications of normal random variables, in the next example, we will present, without proof, a celebrated formula in finance which is constructed based on normal distribution function.

In natural processes, the changes that take place are often the result of a sum of many insignificant random factors. Although the effect of each of these factors, separately, may be ignored, the effect of their sum in general may not. Therefore, the study of distributions of sums of a large number of independent random variables is essential and is the subject of Chapter 11 of the book. In Section 11.5, we have shown that for an immense number of random phenomena, the distribution functions of such sums are approximately normal. In finance, random fluctuations of prices of shares of stocks, or other assets, are a result of an enormity of political, social, economical, and financial factors independently assessed by a large number of traders. For that reason financial scholars and consultants use normal distributions as models for prices of assets in general, and shares of stocks in particular. That is why the *Black-Scholes call option formula*, presented next, is based on normal distribution function.

In finance, a contract that entitles a person to purchase a commodity is called **call option**. To explain the concept of call option, we will present an example. Suppose that Henry purchases, for a price of \$7 per share, the **option** of buying a block of 100 shares in a company at \$80 per share during, say, the next six months. That is, Henry pays \$700 to a *seller* and buys the *right* of purchasing 100 shares in the company at any time before or including the agreed expiration day at \$80 per share. Henry is not obliged to purchase any number of the 100 shares he simply has the option to purchase. However, the premium paid to buy call options is not recovered whether or not he buys any number of the shares. In the language of finance, Henry is the **option holder**, the seller is the **writer**, and the predetermined price per share of \$80, which is valid until the expiration day is called **strike price**. If Henry actually does buy a block of at least one share in the company, then we say that he **exercises** the option.

Suppose that sometime before the expiration day, the price of the shares in the company rise to, say, \$95 per share. If at that time Henry purchases 100 shares for \$80 per share, and sells them immediately for \$95, he makes a profit of $\$9500 - \$8000 - \$700 = \$800$. On the other hand, if the price does not rise by the expiration day, Henry will lose \$700 for having purchased the call option to buy the block of 100 shares. As for the writer, in the latter case, she makes a profit of \$700, whereas in the former case, she will end up selling the block of 100 shares, which is worth \$9500, for only \$8000, losing an opportunity to make $\$1500 - \$700 = \$800$ more.

From this example, it should be evident that determining a fair price for an option is a major problem in *option theory*. The question is whether or not the right to buy a share for a price of \$7 is fair to both Henry and the writer. Until 1973, it was widely believed that the call option should be determined based on the principle assumption that the average return from the stock should be lower than the average return from the option itself. However, in 1973, Fisher Black, an applied mathematician and a financial consultant, and MIT professor Myron Scholes presented a paper with a new approach. They argued, among other points, that all risk will be eliminated if option holders or writers chose two lines of investments with identical returns, buy one and sell the other simultaneously on the expiration day. Based on such assumptions, in terms of normal distribution function, they derived formulas for call options that are extremely popular. To state a version of Black-Scholes' widely used call option formula, note that, since periods remaining to expiration are usually short, those stocks with higher standard deviation will rise above strick price with higher probability. Therefore, they should have higher option values. Interest rate also has an effect on the value of an option. Usually, as interest rate increases, on average, the growth rate of stock price also increases. This should in turn increase the prices of options. Since dividends essentially reduce the stock price, they have negative effect on the values of the call option. For this reason, in the next formula we assume that no dividends are paid before or on the expiration day.

We are now ready to state the *Black-Scholes call option formula* for *European options*, in which, unlike *American options*, the option holder can only exercise on the expiration day. The appearance of the normal distribution function in the following formula has something to do with the fact that the number of upward movements in price of a share of a stock is binomial, and the limiting behavior of binomial is normal.

Black-Scholes Call Option Formula: *In a European call option with standard deviation σ, let K be the strike price, S be the current price, and τ denote the time period over which the option is valid. Suppose that interest is compounded continuously at a rate r. Then the Black-Scholes formula for the value of call option is given by*

$$C = S\Phi(\alpha) - Ke^{-r\tau}\Phi(\alpha - \sigma\sqrt{\tau}),$$

where

$$\alpha = \frac{\ln(S/K) + (r + \sigma^2/2)\tau}{\sigma\sqrt{\tau}},$$

and, as usual, Φ is the distribution function of a standard normal random variable.

To find the Black-Scholes call option value, per share, for the block of 100 shares of the stock Henry is interested in, suppose that the current price of the stock, per share, is \$85 with standard deviation of 18% per year, and the option is valid for 6 months. The strike price is \$80 and the interest rate is 12%. Therefore, $S = 85$, $K = 80$, $\sigma = 0.18$, $r = 0.12$, and $\tau = 6/12 = 1/2$. Hence

$$\alpha = \frac{\ln(85/80) + (0.12 + 0.18^2/2)(1/2)}{0.18\sqrt{1/2}} = 1.01,$$

and thus the value of the call option is

$$\begin{aligned} C &= 85\Phi(1.01) - 80e^{-(0.12)(1/2)}\Phi\big(1.01 - 0.18\sqrt{1/2}\,\big) \\ &= 85\Phi(1.01) - 75.34\Phi(0.88) \\ &= 85(0.8438) - 75.34(0.8106) = 10.65. \quad \blacklozenge \end{aligned}$$

7C THE CAUCHY DISTRIBUTION

A random variable X is called *Cauchy* [*] *with parameter* α, if its probability density function is given by

$$f(x) = \frac{1}{\pi\left[1 + (x-\alpha)^2\right]}, \quad -\infty < x < \infty. \tag{1}$$

As is shown in Example 6.9 of the book, an interesting property of the Cauchy distribution with parameter 0 is that, although it is symmetric with respect to the origin, $E(X)$ does not exist for a Cauchy random variable X; it has no expected value.

Consider the Cauchy density function f given by (1). The graph of f is symmetric about α, and the parameter α is both the unique *median* and the unique *mode* of f. Moreover, the graph of f resembles that of a normal density. For example, the graph of f with $\alpha = 0$ is very similar to that of the probability density function of the standard normal random variable. While it is flatter than the graph of the density of standard normal, it is not as flat as the density of a normal random variable with the mean 0 and variance 2 (see Figure 7a).

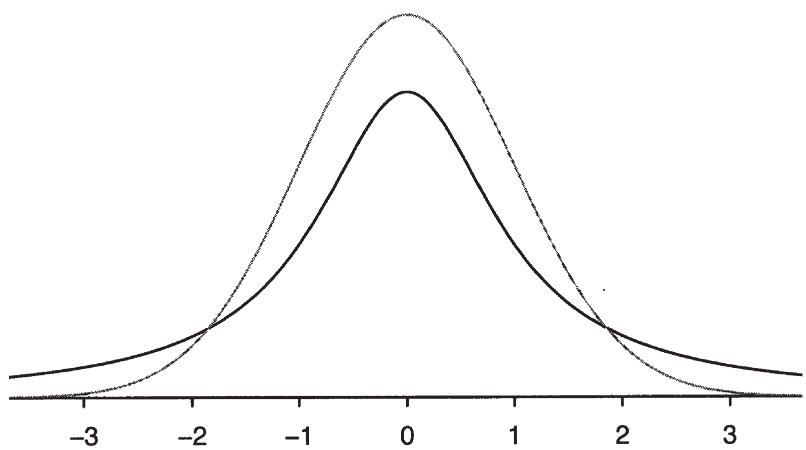

Figure 7a Cauchy density function with $\alpha = 0$.

Example 7h. A vertical mirror freely rotates about 0, the origin of the coordinate system. It projects a horizontal light ray on a wall that has equation $x = 1$. The direction of the reflected ray is a random number. That is, Θ, the angle between the reflected ray and the x-axis, is a uniform random variable over $(-\pi/2, \pi/2)$. Let A be the point $(1, 0)$, B be the intersection of the reflected light ray with the wall, and X be the distance from A to B; that is, $X = \overline{AB}$ (see Figure 7b). Then X is the random variable $\tan\Theta$. To find F, the distribution function of X, note that since Θ is uniform over $(-\pi/2, \pi/2)$,

$$P(\Theta \le t) = \frac{t - (-\pi/2)}{\pi/2 - (-\pi/2)} = \frac{1}{2} + \frac{1}{\pi}t, \quad -\frac{\pi}{2} < t < \frac{\pi}{2}.$$

[*] Augustin Louis Cauchy (1789-1857), born in Paris, was one of the leaders in infusing rigor into Analysis. He is considered to be one of the greatest mathematicians of the 19th century.

Since $y = \tan x$ is an increasing function over the interval $(-\pi/2, \pi/2)$,

$$\begin{aligned} F(t) &= P(X \le t) = P(\tan \Theta \le t) = P(\Theta \le \arctan t) \\ &= \frac{1}{2} + \frac{1}{\pi} \arctan t, \quad -\infty < t < \infty. \end{aligned}$$

The density function of X, f, is obtained by differentiation of F:

$$f(t) = F'(t) = \frac{1}{\pi(1+t^2)}, \quad -\infty < t < \infty.$$

This shows that X is a Cauchy random variable. ♦

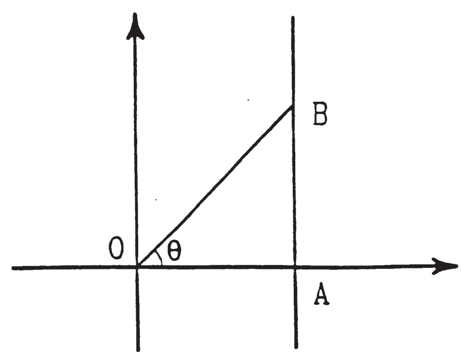

Figure 7b Reflected light ray of Example 7h.

7D THE LOGNORMAL DISTRIBUTION

As we know from Section 7.2 of the book, it is reasonable to assume that such random variables as the height of a man, the weight of a woman, the error made in a measurement, the I. Q. of a person, the growth distribution of plants, animals or their organs, etc. are normally distributed. However, in examples such as the height of a man and the I. Q. of a person, the random variable is strictly positive while the corresponding normal random variable X attains negative values as well. Although in these and similar examples, usually the mean is large and the standard deviation is small making $P(X < 0)$ negligible, still mathematicians and statisticians prefer to use positive random variables as models for positive random variables. For this reason, they have introduced the *lognormal probability density function*, which has been proven to be very useful in such studies as the income distributions, stock market prices, pollutant concentrations, the size of oil fields in America, production data, critical doses of drugs in human beings and animals, etc.

Definition 7a *A random variable Y is called* **lognormal** *if the natural log of Y has a normal distribution. That is, if for some normal random variable X, we have that $Y = e^{X}$.*

Let F be the distribution function of a normal random variable X with parameters μ and σ, and let f be its probability density function. Let $Y = e^{X}$. Then $\ln Y \sim N(\mu, \sigma^2)$, and, G, the distribution function of Y, is calculated as follows.

$$G(y) = P(Y \leq y) = P(e^X \leq y) = P(X \leq \ln y) = F(\ln y).$$

Therefore, g, the density function of Y, which is obtained by differentiating G, is given by

$$g(y) = G'(y) = F'(\ln y) = \frac{1}{y} f(\ln y) = \begin{cases} \dfrac{1}{\sigma y\sqrt{2\pi}} \exp\left[\dfrac{-(\ln y - \mu)^2}{2\sigma^2}\right] & y > 0 \\ 0 & y \leq 0. \end{cases}$$

We will now find the expected value and the standard deviation of a lognormal random variable Y with parameters μ and σ. Let $z = \ln y - \mu$; then $y = e^{z+\mu}$, $dz = \dfrac{1}{y}dy$, and $z \in (-\infty, \infty)$. Thus, for $n \geq 0$,

$$\begin{aligned} E(Y^n) &= \int_0^\infty y^n \cdot \frac{1}{\sigma y\sqrt{2\pi}} \exp\left[\frac{-(\ln y - \mu)^2}{2\sigma^2}\right] dy \\ &= \int_0^\infty y^n \cdot \frac{1}{\sigma\sqrt{2\pi}} \exp\left[\frac{-(\ln y - \mu)^2}{2\sigma^2}\right] \frac{1}{y}\, dy \\ &= \int_{-\infty}^\infty e^{n(z+\mu)} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(\frac{-z^2}{2\sigma^2}\right) dz \\ &= \int_{-\infty}^\infty \frac{1}{\sigma\sqrt{2\pi}} \exp\left[n(z+\mu) - \frac{z^2}{2\sigma^2}\right] dz \\ &= \int_{-\infty}^\infty \frac{1}{\sigma\sqrt{2\pi}} \exp\left[nz + n\mu - \frac{z^2}{2\sigma^2} + \frac{1}{2}\sigma^2 n^2 - \frac{1}{2}\sigma^2 n^2\right] dz \end{aligned}$$

$$= \exp(n\mu + \frac{1}{2}\sigma^2 n^2) \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\left[nz - \frac{z^2}{2\sigma^2} - \frac{1}{2}\sigma^2 n^2\right] dz$$

$$= \exp(n\mu + \frac{1}{2}\sigma^2 n^2) \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\left[\frac{-(z^2 - 2\sigma^2 nz + \sigma^4 n^2)}{2\sigma^2}\right] dz$$

$$= \exp(n\mu + \frac{1}{2}\sigma^2 n^2) \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\left[\frac{-(z - \sigma^2 n)^2}{2\sigma^2}\right] dz$$

$$= \exp(n\mu + \frac{1}{2}\sigma^2 n^2),$$

where the last equality follows since $\frac{1}{\sigma\sqrt{2\pi}} \exp\left[\frac{-(z-\sigma^2 n)^2}{2\sigma^2}\right]$ is the density function of $N(\sigma^2 n, \sigma^2)$, and hence its integral over $(-\infty, \infty)$ is 1. Therefore, for $n \geq 1$,

$$E(Y^n) = \exp\left(n\mu + \frac{1}{2}\sigma^2 n^2\right).$$

This implies that

$$E(Y) = e^{\mu + \frac{1}{2}\sigma^2}, \qquad E(Y^2) = e^{2\mu + 2\sigma^2},$$

and

$$\text{Var}(Y) = E(Y^2) - \big[E(Y)\big]^2 = e^{2\mu+2\sigma^2} - e^{2\mu+\sigma^2} = e^{2\mu+\sigma^2}\left(e^{\sigma^2} - 1\right).$$

Example 7i For a certain spice, one attribute with potential health hazards to the consumers is insect fragments. Although insect fragments are very hard to monitor, studies have shown that the insect fragment count in a bag containing one pound of the spice is lognormal with mean 230 and variance 180. From bags each containing one pound of this spice, one bag is selected at random. What is the probability that its insect fragment count is at most 215?

Solution Let Y be the number of insect fragments in the bag. Then $X = \ln Y \sim N(\mu, \sigma^2)$, where

$$e^{\mu + \frac{1}{2}\sigma^2} = 230, \tag{2}$$

and

$$e^{2\mu+\sigma^2}\left(e^{\sigma^2} - 1\right) = 180. \tag{3}$$

Squaring both sides of (2), we obtain

$$e^{2\mu+\sigma^2} = 52,900, \tag{4}$$

substituting this in (3), we have

$$e^{\sigma^2} - 1 = \frac{180}{52,900} \approx 0.0034.$$

Therefore, $e^{\sigma^2} \approx 1.0034$ and $\sigma^2 \approx \ln 1.0034 \approx 0.0034$, which gives $\sigma \approx 0.0583$. Now substituting $e^{\sigma^2} \approx 1.0034$ in (4), we have $e^{2\mu} \approx 52,900/1.0034 \approx 52,720.75$, which gives

$2\mu \approx \ln 52,720.75 \approx 10.873$ or $\mu \approx 5.44$. Therefore, $X = \ln Y \sim N(5.44, 0.0034)$, and thus the desired probability is obtained as follows.

$$\begin{aligned} P(Y \leq 215) = P(\ln Y \leq \ln 215) = P(X \leq \ln 215) &\approx P(X \leq 5.37) \\ &\approx P\left(\frac{X - 5.44}{0.0583} \leq \frac{5.37 - 5.44}{0.0583}\right) \\ &\approx P(Z \leq -1.20) = 1 - \Phi(1.20) \approx 1 - 0.8849 = 0.1151, \end{aligned}$$

where, as usual, $Z \sim N(0, 1)$. ♦

7E HYPEREXPONENTIAL DISTRIBUTION

Let X_1 and X_2 be exponential random variables with means $1/\lambda_1$ and $1/\lambda_2$, respectively. The random variable Y, defined by

$$Y = \begin{cases} X_1 & \text{with probability } p \\ X_2 & \text{with probability } 1-p, \end{cases}$$

is called **hyperexponential of degree 2** if for all $t \geq 0$ and $s \geq 0$, the event that $X_1 \leq t$ is independent of the event that $X_2 \leq s$. The following example is a general model for hyperexponential random variables.

Example 7j Suppose that a company has two plants. Plant A produces items which have exponential lifetimes with mean $1/\lambda_1$, and, independently of plant A, plant B produces the same type of items but with exponential lifetime distribution with mean $1/\lambda_2$. If $100p$ percent of the total items are produced by plant A and the rest by plant B, and the products are mixed in the market, then Y, the lifetime of a random product of this company in the market, is hyperexponential of order 2. Let F be the distribution function of Y, E be the event that the random product was produced by plant A, and F be the event that it was produced by plant B. By the law of total probability, for $t \geq 0$,

$$\begin{aligned} F(t) = P(Y \leq t) &= P(Y \leq t \mid E)P(E) + P(Y \leq t \mid F)P(F) \\ &= P(X_1 \leq t)P(E) + P(X_2 \leq t)P(F) \\ &= (1 - e^{-\lambda_1 t})p + (1 - e^{-\lambda_2 t})(1-p), \end{aligned}$$

and $F(t) = 0$ for $t < 0$. Let f be the probability density function of Y; we have

$$f(t) = F'(t) = \begin{cases} p\lambda_1 e^{-\lambda_1 t} + (1-p)\lambda_2 e^{-\lambda_2 t} & t \geq 0 \\ 0 & t < 0. \end{cases} \quad \blacklozenge$$

Therefore, as shown in Example 7j, in general, the probability density function, f, of a hyperexponential random variable of order 2 is given by

$$f(t) = \begin{cases} p\lambda_1 e^{-\lambda_1 t} + (1-p)\lambda_2 e^{-\lambda_2 t} & t \geq 0 \\ 0 & t < 0. \end{cases}$$

Hyperexponential random variables are generalized in the following obvious way: For $i = 1, 2, \ldots, n$, let $p_i > 0$, $\sum_{i=1}^n p_i = 1$ and $\lambda_i \geq 0$. The probability density function,

$$g(t) = \begin{cases} \sum_{i=1}^n p_i \lambda_i e^{-\lambda_i t} & t \geq 0 \\ 0 & t < 0, \end{cases}$$

is called **hyperexponential of order** n.